

## CS 200 Project 1: Team5 Requirements Report, Fall 2026

**Header Table:**

| Team Member Name | CWID Number | Crimson Email Address       | Hours Contributed |
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**Task Distribution Table:**

| Team Member Name | Tasks Performed                               | Percentage Contributed |
|------------------|-----------------------------------------------|------------------------|
| Sarah Rechkemmer | Use Case descriptions.                        | 25%                    |
| Ross Craig       | Use Case diagrams, specifically diagrams 3-4. | 25%                    |
| Isaac Kim        | Glossary and detailed written solution.       | 25%                    |
| Jackson Turner   | Use Case diagrams, specifically diagrams 1-2. | 25%                    |

## Detailed Written Solution

Chocoholics Anonymous (ChocAn) is a membership-based platform that gives members access to healthcare providers such as dietitians, internists, and exercise experts. Users pay a monthly fee and receive a card containing their name and a nine-digit member number. Providers use a ChocAn terminal to verify a member's status before providing a service. After the service is completed, the provider enters the date of service, member number, service code, and any optional comments. The system checks the service code, displays the name and fee of the service, and saves a record of the visit. Providers can also request a Provider Directory containing the available services, service codes, and fees. Every Friday at midnight, the system processes the week's service records and creates reports for members, providers, and the accounts-payable manager. It also creates a file containing the information needed to pay each provider. ChocAn operators can add, delete, or update member and provider records, while Acme Accounting Services handles membership payments and updates member statuses each evening.

Our solution is presented in the Glossary, four Use Case Diagrams (Diagram 1: Provider Terminal System, Diagram 2: Provider List, Diagram 3: Accounting Procedure, and Diagram 4: ChocAn Data Center), and the Use Case Descriptions, which are listed in the same order as the diagrams. The actors include ChocAnon Member, ChocAnon Provider, ChocAnon Database Admin, Provider Managers, Data Center Operators, Weekly Automation, and Acme Accounting Services. Some assumptions were made in the use cases. These include the ChocAn Data Center being a part of the system and not modeled as an actor. Weekly Automation represents the scheduled-event trigger as an actor. In our use cases, the displaying of the expiration date for members and notifying members/providers of any updates were assumed optional extensions that could benefit the system design. Extensions contain error statements or unchanged data depending on what has occurred, which assumes that the system can ignore or handle specific error cases.

This project only covers the ChocAn data-processing software. During testing, the provider and manager terminals will be simulated with keyboard input and screen output, and all reports, the Provider Directory, and EFT information will be saved as files instead of being emailed or electronically transferred. Any descriptions containing information that the files or reports will be sent through email is assumed to not be sent during testing and is instead written to a file. The team members are Sarah Rechkemmer, Ross Craig, Isaac Kim, and Jackson Turner. The programming language used will be Java, and the group utilized iMessage to communicate throughout the completion of this project. Sarah Rechkemmer will submit the team's final report.

## Glossary

**Accounts-Payable Manager:** The ChocAn manager who receives a weekly report showing the providers who need to be paid.

**Acme Accounting Services:** The outside company that handles membership payments and updates members' payment statuses each evening.

**Actor:** A person, outside organization, or automated process that interacts with the ChocAn system.

**ChocAn:** Short for Chocoholics Anonymous.

**ChocAn Data Center:** The ChocAn computer facility that stores member, provider, and service records, validates member numbers, and runs the accounting procedure

**ChocAn Data-Processing Software:** The system used to validate members, record services, manage records, and create reports and payment files.

**ChocAnon Database Admin:** An assumed Data Center role that maintains service names, codes, and fees in the Provider Directory

**Comments:** Optional information entered by a provider about a service given to a member.

**Data Center Operators:** ChocAn Data Center staff who, in interactive mode, add, delete, and update member and provider records.

**Date of Service:** The date on which a provider performed a service for a member.

**Electronic Funds Transfer (EFT):** The electronic payment process used to pay providers for their services.

**EFT File:** A file containing the information needed to pay a provider.

**Expiration Date:** An assumed optional member attribute giving the date a membership expires. It is not in the original requirements.

**Fee (Service Fee):** The amount ChocAn pays a provider for a specific service (up to \$999.99).

**Healthcare Provider:** A dietitian, internist, or exercise expert who provides services to ChocAn members.

**Main Accounting Procedure:** The process that runs every Friday at midnight to handle the week's service records.

**Manager:** A ChocAn employee who can request reports and review accounting information.

**Member:** A person who pays a monthly fee to ChocAn and receives access to its healthcare providers.

**Member Card:** A card containing a member's name and nine-digit member number that is used to verify the member's status.

**Member Number:** A unique nine-digit number used to identify a ChocAn member.

**Member Report:** A weekly report listing the services a member received, including the service dates, provider names, and service names.

**Member Status:** A member's current standing in ChocAn. Active or suspended when fees have not been paid for at least a month.

**Notification:** An assumed message sent to a member or provider when their record, payment, or status changes. Written to a file during testing.

**Payment Record:** The Data Center's record of membership fee payments made by a member, updated by Acme.

**Provider Directory:** An alphabetical list of the services, codes, and fees offered by ChocAn.

**Provider Managers:** The ChocAn managers who request reports on demand and receive the accounts-payable summary.

**Provider Number:** A unique nine-digit number used to identify a ChocAn provider.

**Provider Report:** A weekly report listing the services completed by a provider, the number of consultations, and the total amount owed to the provider.

**Provider Terminal:** The device a provider uses to verify members, record services, and request the Provider Directory.

**Service Code:** A unique six-digit number used to identify a particular ChocAn service.

**Service Record:** A saved record of a service that includes the date and time received, date of service, provider number, member number, service code, and optional comments.

**Summary Report:** A report showing the total number of providers, consultations, and fees for the week.

**Suspended:** A member status set when the member has not paid fees for at least a month. Suspended members cannot receive validated services until reinstated.

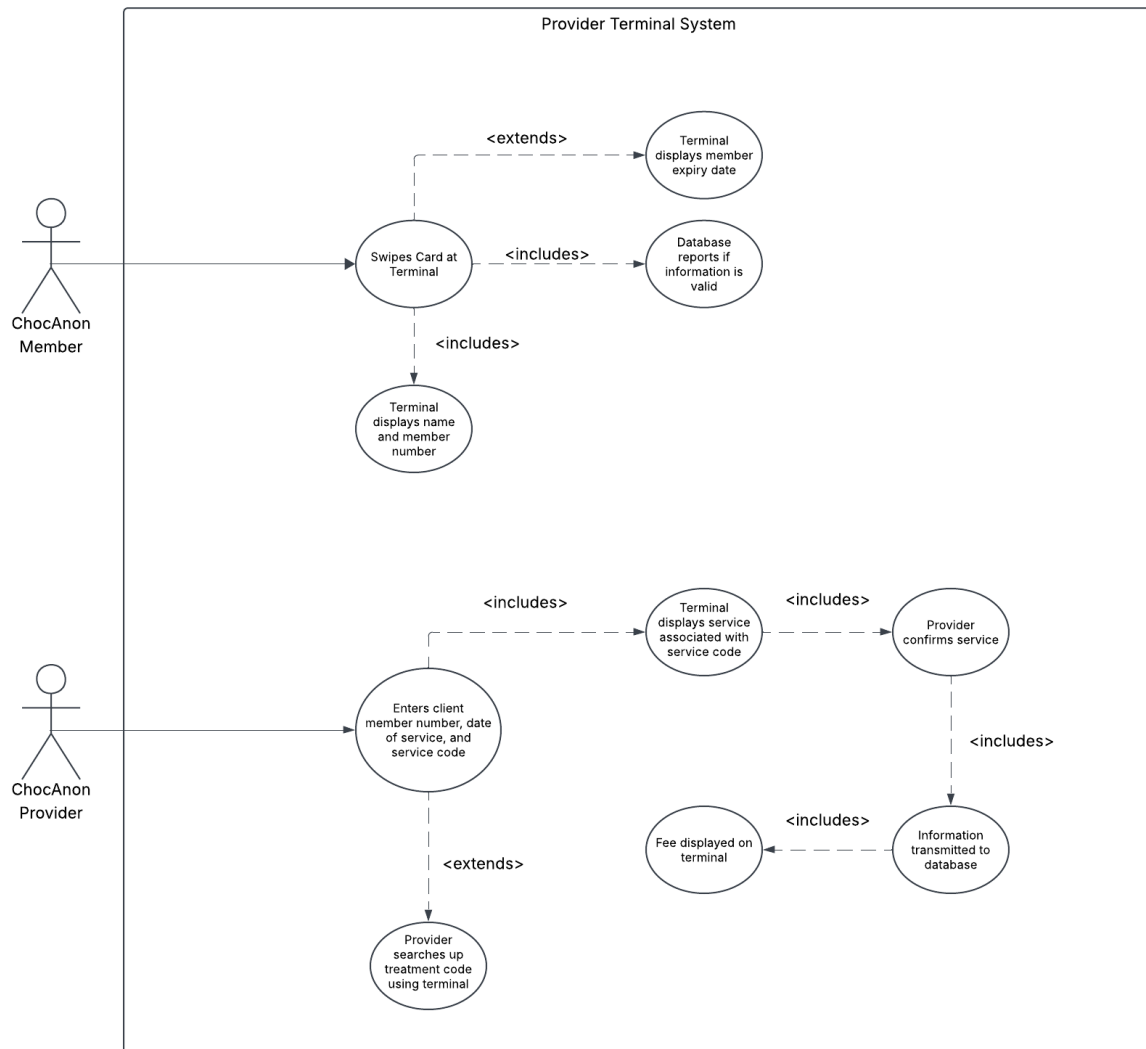
**Validated:** The message shown on the terminal when a member number is valid and the member is active. Other messages are Invalid number and Member suspended.

**Validation:** The process of checking a member's number and status to determine whether the member is eligible to receive a service.

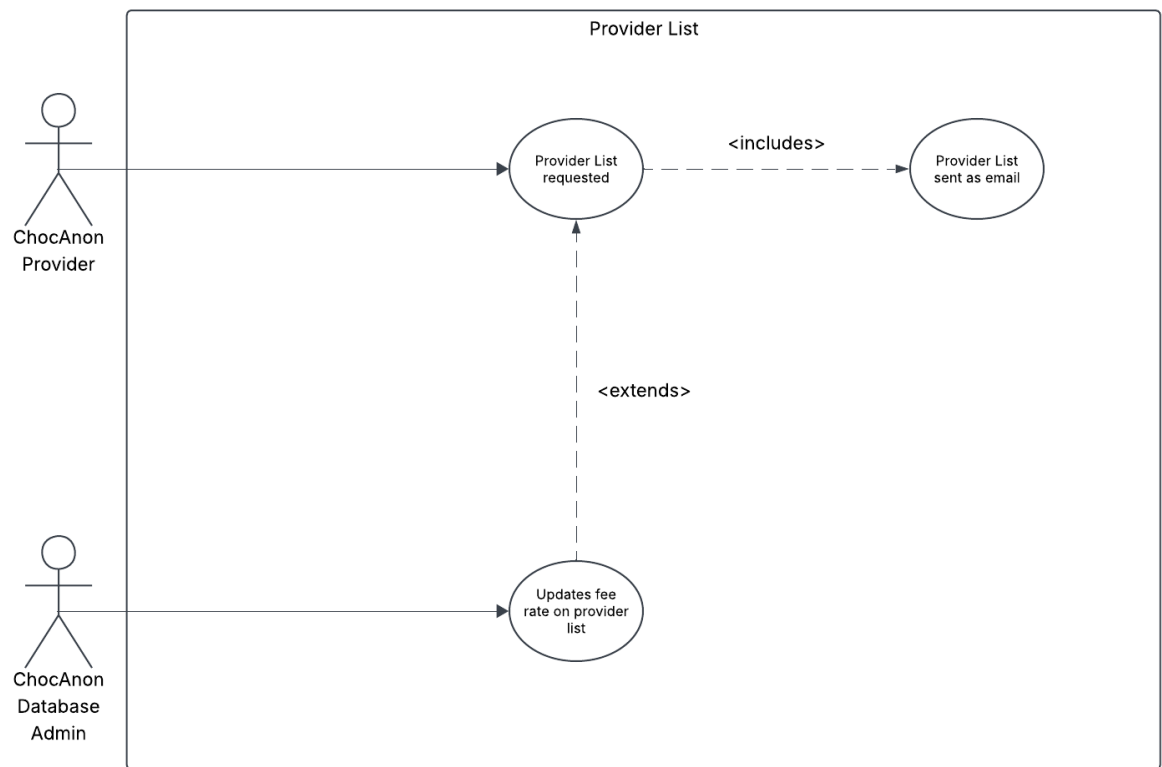
**Verification Form:** The paper form on which a provider records the current date and time, date of service, member name and number, service code, and fee, used to check the weekly payment.

**Weekly Automation:** A scheduled-event actor that triggers the main accounting procedure and EFT creation at midnight every Friday.

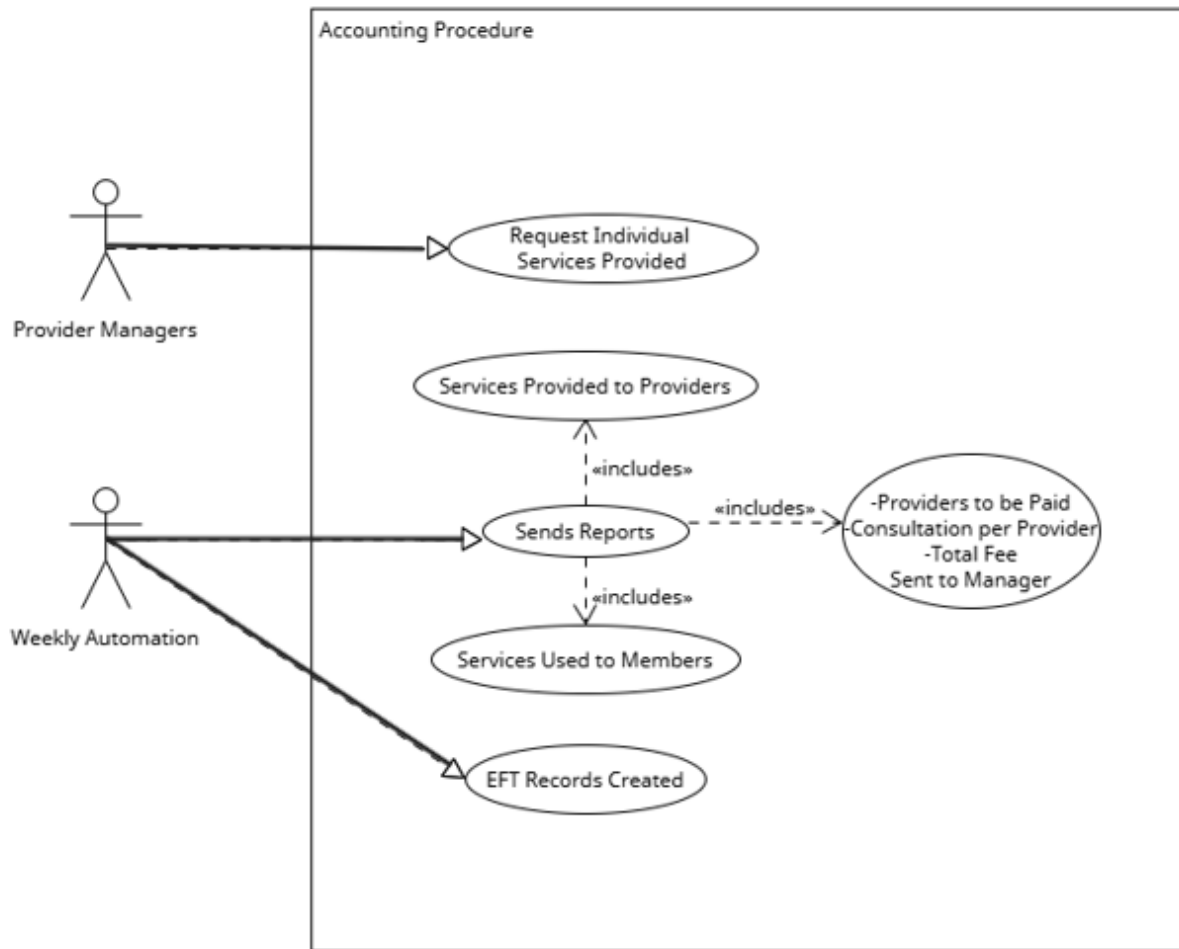
## Use Case Diagrams



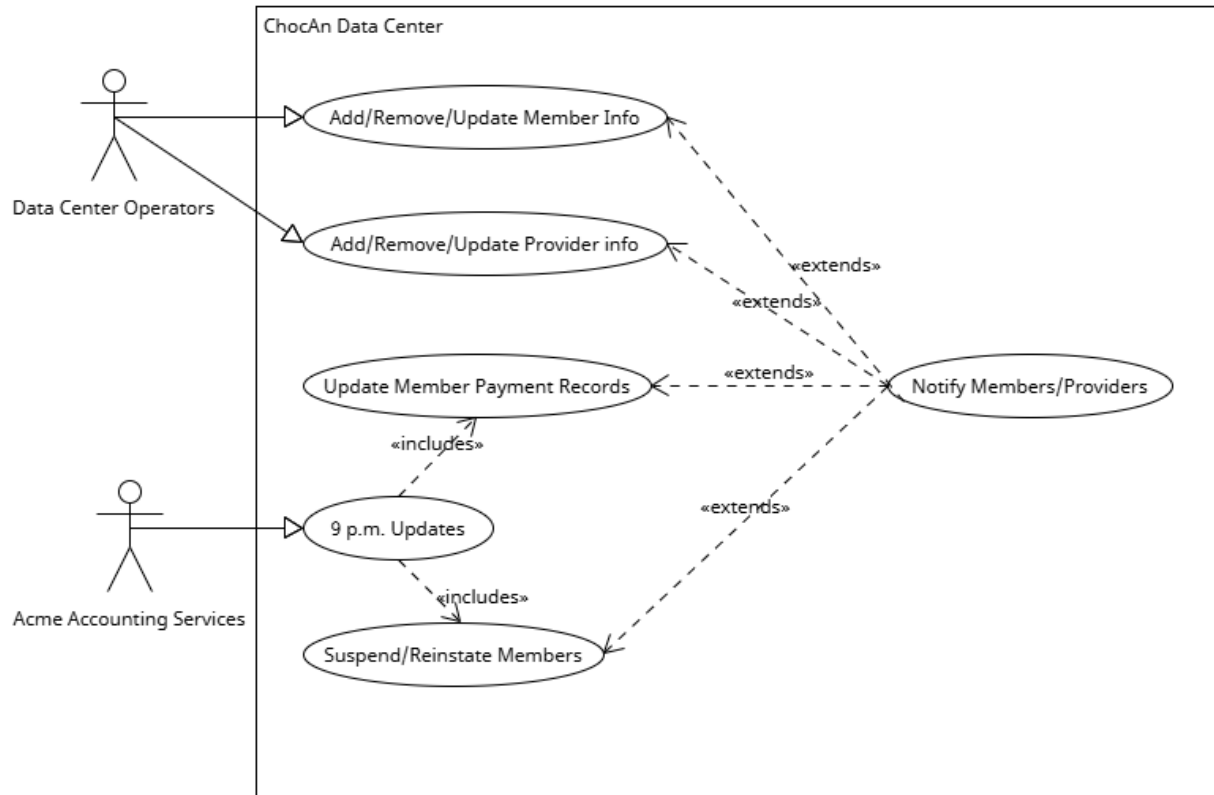
**Diagram 1. Provider Terminal System**



**Diagram 2. Provider List**



**Diagram 3. Accounting Procedure**



**Diagram 4. ChocAn Data Center**

## Use Case Descriptions

### Use Case: Swipes Card at Terminal

Context: A member's card is swiped so that ChocAn can check if the member is valid and not suspended.

Actors: ChocAnon Member

Main Success Scenario:

1. The member hands the card to the provider, who swipes it or keys in the member's number.
2. The terminal sends the member number to the ChocAn Data Center.
3. The system reports whether the information is valid or not to the provider in Database Reports if Information is Valid.
4. The terminal displays the member's name and member number in Terminal Displays Name and Member Number.
5. The terminal shows "Validated."
6. The use case ends and the terminal remains idle until prompted again.

Extensions:

- 1a. The provider keys in the number manually if the card is unreadable.
- 3a. The number is invalid and the terminal displays "Invalid number." The use case ends.
- 3b. The member is suspended because fees are owed and the terminal displays "Member suspended." The use case ends.
- 5a. If the member record has an expiration date, the terminal displays it in Terminal Displays Member Expiry Date.

### Use Case: Database Reports if Information is Valid

Context: The Data Center checks the member number and status and returns the result.

Actors: ChocAnon Member

Main Success Scenario:

1. The Data Center receives the member number.
2. The center looks up the member record.
3. It confirms the number exists and is active.
4. It returns "Validated" to the terminal.
5. The use case ends and is returned to Swipes Card at Terminal.

Extensions:

- 2a. No matching number provided and “Invalid number” is displayed.
- 3a. Member status is suspended and “Member suspended” is displayed.

### **Use Case: Terminal Displays Name and Member Number**

Context: The terminal shows whose card was read to confirm identities.

Actors: ChocAnon Member

Main Success Scenario:

- 1. The Data Center returns the member’s name and number.
- 2. The terminal displays the information.
- 3. The provider identifies the member.
- 4. The use case ends and returns to Swipes Card at Terminal.

Extensions:

- 3a. The information does not match, and the provider swipes the card again.

### **Use Case: Terminal Displays Member Expiry Date**

Context: Assumption that the terminal will display the date when the member’s membership expires.

Actors: ChocAnon Member

Main Success Scenario:

- 1. After card is validated, the system checks for an expiration date.
- 2. The Data Center retrieves the information of the member.
- 3. The terminal displays the member information and the expiration date.
- 4. The use case ends and returns to Swipes Card at Terminal.

Extensions:

- 1a. There is no expiration date to be displayed to the terminal.

### **Use Case: Enters Client Member Number, Date of Service, and Service Code**

Context: After providing a service, the provider bills ChocAn by entering the details.

Actors: ChocAnon Provider

Main Success Scenario:

- 1. The provider turns on the terminal and enters their provider number.

2. The provider passes the card through the reader or keys in the member number, and “Validated” appears on the terminal.
3. The provider enters the date of service in the format MM-DD-YYYY.
4. The provider enters the six-digit service code for what was provided.
5. The terminal displays the service for that specific code in Terminal Displays the Service Associated with the Service Code.
6. The provider confirms the service code in Provider Confirms Service.
  - 6.1. The provider can enter an optional comment of 100-characters.
7. The information is sent to the database and the fee of the service is displayed in Information Transmitted to Database and Fee Displayed on Terminal.
8. The use case ends and the terminal remains idle until prompted again.

Extensions:

- 2a. The “Validated” message does not appear and the member is invalid or suspended. The use case ends.
- 3a. The date entered is in an invalid format, and the provider must input it again.
- 4a. The provider does not know the code and uses Provider Searches up Treatment Code Using Terminal.
- 5a. The code for the service does not exist and an error message is displayed.
- 5b. The provider reenters the code for the service because it is incorrect, invalid, or does not exist.
- 6.1a. The provider chooses not to enter a comment for the service provided.

**Use Case: Terminal Displays the Service Associated with the Service Code**

Context: The terminal shows the service name so the provider can determine if the code is correct.

Actors: ChocAnon Provider

Main Success Scenario:

1. The system finds the service associated with the code.
2. The terminal displays the service name up to 20 characters.
3. The provider checks the name against the service performed.
4. The provider verifies the service in Provider Confirms Service.
5. The use case ends and returns to Enters Client Member Number, Date of Service, and Service Code.

Extensions:

1a. The code does not exist and an error message is displayed. The provider reenters the code and tries again.

3a. The name does not match the service provided and thus the provider does not verify it.

### **Use Case: Provider Confirms Service**

Context: The provider verifies that the service displayed is the one provided and performed.

Actors: ChocAnon Provider

Main Success Scenario:

1. The terminal asks the provider to verify that the service occurred.
2. The provider confirms the service.
3. The system transmits information to database that service was approved in Information Transmitted to Database.
4. The use case ends and returns to Terminal Displays Service Associated with Service Code.

Extensions:

- 2a. The provider declines and does not verify the service. The service is discarded and a record of the service is not saved.

### **Use Case: Information Transmitted to Database**

Context: The confirmed billing information is saved by the Data Center and recorded to a disk.

Actors: ChocAnon Provider

Main Success Scenario:

1. The system creates a service record with the current date and time, date of service, provider number, member number, service code, and any comments the provider wrote.
2. The system writes the record to disk.
3. The system looks up the fee and displays it in Fee Displayed on Terminal.
4. The use case ends and returns to Provider Confirms Service.

Extensions:

- 2a. A disk error occurs and the system does not record the entry. The provider can retry.

**Use Case: Fee Displayed on Terminal**

Context: Shows the provider the fee that ChocAn will pay for the service provided.

Actors: ChocAnon Provider

Main Success Scenario:

1. The system finds the fee for the associated service code.
2. The terminal displays the fee.
3. The provider writes the fee with the current date and time, date of service, member name, number, and service code on the verification form.
4. The provider verifies the fee.
5. The use case ends and returns to Information Transmitted to Database.

Extensions:

- 4a. The provider does not verify the service fee and contacts ChocAn.

**Use Case: Provider Searches up Treatment Code using Terminal**

Context: The provider can look up the six-digit code for the service through the Provider Directory.

Actors: ChocAnon Provider

Main Success Scenario:

1. The provider chooses to look up a service code through the directory.
2. The provider requests the Provider Directory in Provider List Requested.
3. The terminal displays the list of services and their respective codes.
4. The provider identifies the service code.
5. The use case ends and the provider keys in the service code in Enters Client Member Number, Date of Service, and Service Code.

Extensions:

- 3a. The service is not in the list, and the provider must contact ChocAn.

**Use Case: Provider List Requested**

Context: A provider asks for the Provider Directory, which is an alphabetical list of services, their codes, and the fees associated with them.

Actors: ChocAnon Provider

Main Success Scenario:

1. The provider requests the Provider Directory.

2. The system retrieves all the services with their respective codes and fees.
3. The system displays them alphabetically by service name.
4. The list is emailed to the provider in Provider List Sent as Email.
5. The use case ends.

Extensions:

- 1a. The provider cancels the request.
- 1b. A service fee has changed and Updates Fee rate on Provider List is performed first.
- 2a. The system cannot retrieve all services and an error occurs. The system must be retried.

### **Use Case: Provider List Sent as Email**

Context: The requested Provider Directory is delivered to the provider as an email attachment.

Actors: ChocAnon Provider

Main Success Scenario:

1. The system formats the list.
2. The list is written to a file.
3. The list is sent as an email to the provider.
4. The use case ends and returns to Provider List Requested.

Extensions:

- 2a. The file cannot be created, and an error message is displayed, and the provider must try again.
- 3a. The email cannot be sent, and an error message is displayed, and the provider can try again.

### **Use Case: Updates Fee Rate on Provider List**

Context: A database administrator updates the fees associated with the services.

Actors: ChocAnon Database Admin

Main Success Scenario:

1. The database administrator selects the service.
2. The system displays the current service name, code, and fee.
3. The database admin enters the new service fee and checks that it is no more than \$999.99.

4. The system saves the update.
5. The use case ends and returns to Provider List Requested.

Extensions:

- 1a. Service is not found or invalid, and the admin contacts ChocAn or tries again.
- 3a. The service fee is more than \$999.99 and invalid, and an error is displayed until the admin reenters the fee.

### **Use Case: Sends Reports**

Context: At midnight on Fridays, the system reads the week's service records and creates all weekly reports.

Actors: Weekly Automation

Main Success Scenario:

1. At midnight on Friday, weekly automation starts working on the accounting procedure.
2. The system reads the week's file of services provided.
3. The provider reports are created in Services Provided to Providers.
4. The member reports are created in Services Used to Members.
5. A summary report is created for the manager, including providers to be paid, consultations per provider, and total fees owed in Providers to be Paid, Consultation per Provider, and Total Fee Sent to Manager.
6. The reports are written to a file and sent out.
7. The use case ends.

Extensions:

- 2a. No services were provided throughout the week and data remains unchanged.

### **Use Case: Services Provided to Providers**

Context: Each provider who provides a service during the week gets a report of the services they provided.

Actors: Weekly Automation

Main Success Scenario:

1. The system selects the providers who billed throughout the week.
2. For each report, it lists the name, number, address, city, state, and ZIP.
3. For each service, it lists the date of service, date and time received, member name, member number, service code, and fee in the order received.

4. The report contains a summary of the number of consultations and the total fee for the week.
5. Report is written to a file named with the provider's name followed by the report date.
6. The use case ends and returns to Sends Reports.

Extensions:

- 1a. No providers were billed and therefore no report is generated. The data remains unchanged.

### **Use Case: Services Used to Members**

Context: Each member who received a service throughout the past week gets a report of the services provided to them.

Actors: Weekly Automation

Main Success Scenario:

1. The system selects members who received services throughout the week.
2. For each report, it lists the name, number, address, city, state, and ZIP code.
3. For each service, it lists the date of service, provider name, and service name, sorted by the date.
4. It writes the report to a file named with the member's name and date.
5. The use case ends and returns to Sends Reports.

Extensions:

- 1a. No members received services throughout the week, and therefore no report is generated. The data remains unchanged.

### **Use Case: Providers to be Paid, Consultation per Provider, and Total Fee Sent to Manager**

Context: The manager for the accounts payable receives a weekly summary of who needs be paid.

Actors: Weekly Automation

Main Success Scenario:

1. The system lists the total amount of providers to be paid throughout the past week.
2. For each provider, it shows the total number of consultations and total fees.
3. Then, the total number of providers who provided services, the total number of consultations, and the overall fees are printed.
4. The summary is displayed to the manager.

5. The use case ends and returns to Sends Reports.

Extensions:

- 1a. No provider needs to be paid, and no services were accumulated throughout the week. The data remains unchanged and the report displays zero providers, consultations, or fees.

### **Use Case: EFT Records Created**

Context: The system creates the payment information used by the banking system.

Actors: Weekly Automation

Main Success Scenario:

1. The system reads the provider's weekly totals.
2. For each provider, the provider's name, number, and fee amount is written to the EFT file.
3. The system is complete.
4. The use case ends.

Extensions:

- 1a. No provider is owed money, and the system ends as no data is changed.

### **Use Case: Request Individual Services Provided**

Context: A manager can run weekly reports on demand at any time during the week.

Actors: Provider Managers

Main Success Scenario:

1. The manager chooses a report to run.
2. The system reads the current service records.
3. The system produces the requested report.
4. The manager views the report and saves it as a file.
5. The use case ends.

Extensions:

- 1a. The manager does not select a report and the operation is ended.
- 2a. The records for the report do not exist and the use case ends.

### **Use Case: Add/Remove/Update Member Info**

Context: In interactive mode, the Data Center Operator can add, delete, or update member information.

Actors: Data Center Operators

Main Success Scenario:

1. The operator chooses add, remove, or update.
2. The operator enters the member data.
3. The system checks and saves the update.
4. The system confirms the update to the operator.
5. The use case ends.

Extensions:

- 1a. The operator cancels the operation.
- 3a. The member number already exists when attempting to add a member. An error occurs and the operator must retry.
- 3b. The member number is not found or does not exist. An error occurs and the operator must retry.
- 3c. The member's data is invalid and displays an error for the operator to try again.
- 4a. The system triggers Notify Members/Providers.

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Context: In interactive mode, the Data Center Operator can add, delete, or update provider information.

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Main Success Scenario:

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3. The system checks and saves the update.
4. The system confirms the update to the operator.
5. The use case ends.

Extensions:

- 1a. The operator cancels the operation.
- 3a. The provider number already exists when attempting to add a provider. An error occurs and the operator must retry.
- 3b. The provider number is not found or does not exist. An error occurs and the operator must retry.

3c. The provider's data is invalid and displays an error for the operator to try again.

4a. The system triggers Notify Members/Providers.

### **Use Case: 9 P.M. Updates**

Context: Every evening at 9 PM, Acme updates ChocAn's membership records with payment and status changes.

Actors: Acme Accounting Services

Main Success Scenario:

1. At 9 PM, Acme sends the updated membership information.
2. The system matches each entry to a member record.
3. The system updates the payment records in Update Member Payment Records.
4. The system suspends or reinstates members based on the payment in Suspend/Reinstate Members.
5. The system saves records.
6. The use case ends.

Extensions:

- 1a. Acme's data is not received and the records remain unchanged.
- 2a. Member data is invalid and not found to match, displaying an error. The system will skip the entry and record the error presented.

### **Use Case: Update Member Payment Records**

Context: Acme records membership fee payments received from members in the ChocAn Data Center's member records.

Actors: Acme Accounting Services

Main Success Scenario:

1. The system receives payment from Acme.
2. The system matches the entry to a member record by their number.
3. The system updates the member's payment record with the payment and date.
4. The record is saved.
5. The use case ends and returns to 9 P.M. Updates.

Extensions:

- 2a. No member record matches the number and an error is produced.
- 3a. Payment data is invalid and the system leaves it unchanged and an error is produced.

4a. Payment record changes and the system triggers Notify Members/Providers.

### **Use Case: Suspend/Reinstate Members**

Context: Changes the member status based on fee payments.

Actors: Acme Accounting Services

Main Success Scenario:

1. The system checks each member's payment status.
2. If a member has not paid for at least a month, it sets the status to suspended.
3. If a suspended member has now paid what is owed, the status is set to active.
4. The use case ends and returns to 9 P.M. Updates.

Extensions:

- 1a. Status is already correct for a member.
- 2a. A member is notified of the change through Notify Members/Providers.
- 3a. A member is notified of the change through Notify Members/Providers.

### **Use Case: Notify Members/Providers**

Context: An assumption that a member or provider is notified when their record or status is changed for their information, payment records, or suspension/reinstating.

Actors: Data Center Operators, Acme Accounting Services

Main Success Scenario:

1. The system detects a change to a member's or provider's record, payment, or status.
2. The system creates a notice for the selected member/provider.
3. The notice is sent to the member/provider.
4. The use case ends.

Extensions:

- 3a. The notice cannot be delivered and the system must try again.

## References

For this project, the Term Project: Chocoholics Anonymous pgs 627-629 documents were referenced and utilized. Additionally, UMLetino was used to create the Use Case Diagrams.

AI Note: For this assignment's preparation, the team utilized Claude, a language model created by Anthropic. Within this assignment, Claude was utilized to help check for grammar, logical errors, or terms missed within the Glossary. Additionally, Claude helped clarify how to include "The case ends and returns to \_\_\_\_" statements and extension errors in the Case Descriptions, as well as explain in greater detail how to construct a Use Case Diagram.